

Homework 7

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Problem 1

(a) We note that

$$\langle 0 | e^{ik\hat{x}} | 0 \rangle = \int_{-\infty}^{\infty} \langle 0 | e^{ik\hat{x}} | x \rangle \langle x | 0 \rangle dx.$$

Since $|x\rangle$ is an eigenstate of $\hat{x}$, the exponential of $\hat{x}$ acts on this by

$$= \int_{-\infty}^{\infty} \langle 0 | e^{ikx} | x \rangle \langle x | 0 \rangle dx = \int_{-\infty}^{\infty} e^{ikx} \langle 0 | x \rangle \langle x | 0 \rangle dx.$$

One can prove this by writing out the Taylor expansion and factoring in $|x\rangle$. Since $\langle x | 0 \rangle = \psi_0(x)$ for ψ_0 the time-independent wavefunction in the ground state of the oscillator, this equals

$$= \int_{-\infty}^{\infty} e^{ikx} |\psi_0(x)|^2 dx.$$

This is the Fourier transform of the probability distribution of the ground state. The ground state probability density is

$$|\psi_0(x)|^2 = \frac{1}{\sqrt{2\pi x_0^2}} e^{-x^2/2x_0^2},$$

where

$$x_0^2 = \frac{\hbar}{2m\omega}.$$

This is just a gaussian, and the Fourier transform of a gaussian is just

$$\int_{-\infty}^{\infty} e^{ikx} |\psi_0(x)|^2 dx = e^{-k^2 x_0^2 / 2}.$$

It suffices to show that $\langle 0 | \hat{x}^2 | 0 \rangle = x_0^2$. Note that

$$\hat{x} = x_0 (\hat{a} + \hat{a}^\dagger).$$

We can use the properties that $\hat{a}|0\rangle = 0$, $\langle 0|\hat{a}^\dagger = 0$, and $[\hat{a}, \hat{a}^\dagger] = 1$ to show

$$\begin{aligned} \langle 0 | x_0^2 (\hat{a} + \hat{a}^\dagger)^2 | 0 \rangle &= x_0^2 (\langle 0 | \hat{a}^2 | 0 \rangle + \langle 0 | (\hat{a}^\dagger)^2 | 0 \rangle + \langle 0 | \hat{a} \hat{a}^\dagger | 0 \rangle + \langle 0 | \hat{a}^\dagger \hat{a} | 0 \rangle) \\ &= x_0^2 (\langle 0 | \hat{a} \hat{a}^\dagger | 0 \rangle + \langle 0 | \hat{a}^\dagger \hat{a} | 0 \rangle) = \langle 0 | \hat{a}^\dagger \hat{a} + 1 | 0 \rangle = x_0^2 (\langle 0 | \hat{a}^\dagger \hat{a} | 0 \rangle + \langle 0 | 0 \rangle) = x_0^2. \end{aligned}$$

Therefore, the exponential we found is

$$= \exp \left(-\frac{k^2}{2} \langle 0 | \hat{x}^2 | 0 \rangle \right).$$

(b) By all of the same logic, but now in momentum space,

$$\begin{aligned}\langle 0|e^{ix\hat{k}}|0\rangle &= \int_{-\infty}^{\infty} \langle 0|e^{ix\hat{k}}|0\rangle dp = \int_{-\infty}^{\infty} \langle 0|e^{ixp/\hbar}|p\rangle \langle p|0\rangle dp \\ &= \int_{-\infty}^{\infty} e^{ixp/\hbar} |\psi_0(p)|^2 dp.\end{aligned}$$

Because the wavefunction in the momentum space is the Fourier transform of the wavefunction in position space, and the Fourier transform of a gaussian is a gaussian, we can write

$$|\psi_0(p)|^2 = \frac{1}{2\pi k_0^2} e^{-p^2/2p_0^2}, \quad k_0^2 = \frac{m\omega}{2\hbar}.$$

The Fourier transform is thus

$$= \exp\left(-\frac{x^2 k_0^2}{2\hbar^2}\right).$$

It suffices now to show that $k_0^2 = \langle 0|\hat{p}^2|0\rangle$. Note that

$$\hat{p} = i\sqrt{\frac{m\omega\hbar}{2}} (\hat{a}^\dagger - \hat{a}) \implies \hat{k} = \frac{1}{\hbar}\hat{p} = i\sqrt{\frac{m\omega}{2\hbar}} (\hat{a}^\dagger - \hat{a}) = ik_0 (\hat{a}^\dagger - \hat{a}).$$

Since $\hat{a}|0\rangle$ and $\langle 0|\hat{a}^\dagger$ are both 0, we can cancel most of the terms as before. Using the commutator expansion once again, we have that

$$\langle 0|\hat{k}^2|0\rangle = -i^2 k_0^2 \langle 0|\hat{a}\hat{a}^\dagger|0\rangle = k_0^2.$$

Problem 2

(a) We start by solving the Schrödinger equation in position space. The time-independent wavefunction must be of the form $\psi(x) = Ae^{\kappa x} + Be^{-\kappa x}$ for some real $\kappa > 0$. The solutions are not imaginary because we are working with a bound state, which we take to have negative energy. For $\psi(x)$ to be normalizeable, it must go to 0 as $x \rightarrow \pm\infty$, meaning

$$\psi(x) = \begin{cases} Ae^{\kappa x}, & x \leq 0 \\ Be^{-\kappa x}, & x \geq 0 \end{cases}.$$

Because both piecewise components must agree at $x = 0$, we see that $A = B$. We determine A from normalizeability:

$$\begin{aligned}\int_{-\infty}^{\infty} |\psi(x)|^2 dx &= A^2 \left(\int_{-\infty}^0 e^{2\kappa x} dx + \int_0^{\infty} e^{-2\kappa x} dx \right) = A^2 \left(\frac{1}{2\kappa} e^{2\kappa x} \Big|_{-\infty}^0 - \frac{1}{2\kappa} e^{-2\kappa x} \Big|_0^{\infty} \right) \\ &= A^2 \left(\frac{1}{2\kappa} - 0 + \frac{1}{2\kappa} - 0 \right) = \frac{A^2}{\kappa} = 1 \implies A = \sqrt{\kappa}.\end{aligned}$$

The wavefunction is thus $\sqrt{\kappa}e^{-\kappa|x|}$.

Now we Fourier transform into k -space using the Fourier transform into momentum space by relating k and momentum. Using the convention $\langle k|x\rangle = \frac{e^{-ikx}}{\sqrt{2\pi}}$,

$$\langle k|\psi_b\rangle = \int_{-\infty}^{\infty} \langle k|x\rangle \langle x|\psi_b\rangle dx = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-ikx} \psi(x) dx = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-ikx} \sqrt{\kappa} e^{-\kappa|x|} dx$$

$$\begin{aligned}
&= \sqrt{\frac{\kappa}{2\pi}} \left(\int_{-\infty}^0 e^{(\kappa-ik)x} dx + \int_0^{\infty} e^{-(\kappa+ik)x} dx \right) \\
&= \sqrt{\frac{\kappa}{2\pi}} \left(\left[\frac{e^{(\kappa-ik)x}}{\kappa-ik} \right]_{-\infty}^0 + \left[\frac{e^{-(\kappa+ik)x}}{-(\kappa+ik)} \right]_0^{\infty} \right) = \sqrt{\frac{\kappa}{2\pi}} \left(\frac{1}{\kappa-ik} - \frac{1}{-(\kappa+ik)} \right) \\
&= \sqrt{\frac{\kappa}{2\pi}} \frac{\kappa+ik+\kappa-ik}{(\kappa-ik)(\kappa+ik)} = \sqrt{\frac{\kappa}{2\pi}} \frac{2\kappa}{\kappa^2+k^2}.
\end{aligned}$$

(b) The wavefunction $\psi_2(x)$ in position space is

$$\psi_2(x) = \sqrt{\frac{2}{a}} \sin \frac{2\pi x}{a}, \quad x \in [0, a],$$

and it vanishes outside the box. Fourier transforming to momentum space:

$$\psi_2(p) = \int_{-\infty}^{\infty} \langle p | x \rangle \langle x | \psi_2 \rangle dx = \int_0^a \langle p | x \rangle \psi_2(x) dx.$$

Using the relation $\langle p | x \rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{ipx/\hbar}$, we have

$$= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\hbar}} e^{ipx/\hbar} \sqrt{\frac{2}{a}} \sin \frac{2\pi x}{a} dx = \frac{1}{\sqrt{\pi a \hbar}} \int_{-\infty}^{\infty} e^{ipx/\hbar} \sin \frac{2\pi x}{a} dx.$$

We use Euler's formula to write $\sin \theta = \frac{1}{2i}(e^{i\theta} - e^{-i\theta})$. Applying this here yields

$$= \frac{1}{2i\sqrt{\pi a \hbar}} \left(\int_{-\infty}^{\infty} e^{i(2\pi/a-p/\hbar)x} dx - e^{-i(2\pi/a+p/\hbar)x} dx \right).$$

I did not feel like typing up all the work so this simplifies down to

$$\psi_2(p) = \frac{2\hbar\sqrt{\pi a \hbar}}{4\pi^2\hbar^2 - p^2a^2} \left(1 - e^{-ipa/\hbar} \right).$$

The probability density is then

$$|\psi_2(p)|^2 = \frac{4\pi a \hbar^3}{(4\pi^2\hbar^2 - p^2a^2)^2} \left| 1 - e^{-ipa/\hbar} \right|^2.$$

You have to use

$$\left| 1 - e^{-i\theta} \right|^2 = (1 - \cos \theta)^2 + \sin^2 \theta = 1 - 2 \cos \theta + \cos^2 \theta + \sin^2 \theta = 2(1 - \cos \theta) = 4 \sin^2 \frac{\theta}{2}.$$

This yields

$$|\psi_2(p)|^2 = \frac{16\pi a \hbar^3}{(4\pi^2\hbar^2 - p^2a^2)^2} \sin^2 \frac{pa}{2\hbar}.$$

Therefore,

$$dW(p) = \frac{16\pi a \hbar^3}{(4\pi^2\hbar^2 - p^2a^2)^2} \sin^2 \frac{pa}{2\hbar} dp.$$

Problem 3

i didnt do this one

Problem 4

(a) We start with

$$\hat{x}_H = e^{i\hat{H}t/\hbar} \hat{x} e^{i\hat{H}t/\hbar}, \quad \hat{x}_H(0) = \hat{x}(0) = \Delta_x(\hat{a} + \hat{a}^\dagger).$$

We can then write

$$\hat{x}_H(t) = \Delta_x \left(e^{i\hat{H}t/\hbar} \hat{a} e^{-i\hat{H}t/\hbar} + e^{i\hat{H}t/\hbar} \hat{a}^\dagger e^{-i\hat{H}t/\hbar} \right).$$

Note that the terms inside the parentheses are precisely the Heisenberg picture annihilation operators, so

$$\hat{x}_H(t) = \Delta_x \left(\hat{a}_H(t) + \hat{a}_H^\dagger(t) \right).$$

To find the annihilation operators, note that the oscillator Hamiltonian in the Schrödinger picture can be written as

$$\hat{H} = \hbar\omega \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right).$$

Since the Hamiltonian is time-independent, we can write $\exp(i\hat{H}t/\hbar)$ as U , which will simplify notation considerably. We can use the Heisenberg equation of motion

$$i\hbar \frac{d\hat{a}_H}{dt} = [\hat{a}_H, \hat{H}_H] = [U\hat{a}U^\dagger, U\hat{H}U^\dagger] = U[\hat{a}, \hat{H}]U^\dagger.$$

We can compute the commutator in the Schrödinger picture:

$$\begin{aligned} [\hat{a}, \hat{H}] &= \hbar\omega \left[\hat{a}, \hat{a}^\dagger \hat{a} + \frac{1}{2} \right] = \hbar\omega \left[\hat{a}, \hat{a}^\dagger \hat{a} \right] + \hbar\omega \left[\hat{a}, \frac{1}{2} I \right] = \hbar\omega \left[\hat{a}, \hat{a}^\dagger \hat{a} \right] \\ &= \hbar\omega \left(\hat{a} \hat{a}^\dagger \hat{a} - \hat{a}^\dagger \hat{a}^2 \right) = \hbar\omega \left[\hat{a}, \hat{a}^\dagger \right] \hat{a} = \hbar\omega \hat{a}. \end{aligned}$$

We then have

$$\frac{d\hat{a}_H}{dt} = -i\omega \hat{a}_H(t).$$

The differential equation is solved by

$$\hat{a}_H(t) = \hat{a}_H(0) e^{-i\omega t} = \hat{a} e^{-i\omega t}.$$

We then have

$$\hat{a}_H^\dagger(t) = \hat{a}^\dagger e^{i\omega t}.$$

We then compute

$$\hat{x}_H(t) = \Delta_x \left(\hat{a} e^{-i\omega t} + \hat{a}^\dagger e^{i\omega t} \right).$$

Substituting this into the definition for $g_x(t, 0)$, we obtain

$$g_x(t, 0) = \Delta_x^2 \left\langle n \left| \left(\hat{a} e^{-i\omega t} + \hat{a}^\dagger e^{i\omega t} \right) \left(\hat{a} + \hat{a}^\dagger \right) \right| n \right\rangle.$$

Expanding this out:

$$= \Delta_x^2 \langle n | \left(\hat{a}^2 e^{-i\omega t} + (\hat{a}^\dagger)^2 e^{i\omega t} + \hat{a} \hat{a}^\dagger e^{-i\omega t} + \hat{a}^\dagger \hat{a} e^{i\omega t} \right) | n \rangle.$$

We of course have that

$$\langle n | \hat{a} \hat{a} | n \rangle = \langle n | n + 2 \rangle = 0, \quad \langle n | \hat{a}^\dagger \hat{a}^\dagger | n \rangle = \langle n | n - 2 \rangle = 0.$$

Keeping the other terms only yields

$$g_x(t, 0) = \Delta_x^2 e^{-i\omega t} \langle n | \hat{a} \hat{a}^\dagger | n \rangle + \Delta_x^2 e^{i\omega t} \langle n | \hat{a}^\dagger \hat{a} | n \rangle = \Delta_x^2 ((n+1)e^{-i\omega t} + ne^{i\omega t}).$$

To compute $g_x(0, t)$, we note that

$$g_x(0, t) = \langle n | \hat{x}_H(0) \hat{x}_H(t) | n \rangle = |n\rangle^\dagger \hat{x}_H^\dagger(0) \hat{x}_H^\dagger(t) \langle n |^\dagger = (\langle n | \hat{x}_H(t) \hat{x}_H(0) | n \rangle)^\dagger = g_x^\dagger(t, 0).$$

We therefore have

$$g_x(0, t) = \Delta_x^2 ((n+1)e^{i\omega t} + ne^{-i\omega t}).$$

These are not going to be equal in general since they are complex conjugates and are not purely real unless $\omega = n\pi$.

(b) The symmetric correlation function is given by

$$\begin{aligned} g_x(t) &= \frac{1}{2} (g_x(t, 0) + g_x(0, t)) = \frac{\Delta_x^2}{2} ((n+1)(e^{i\omega t} + e^{-i\omega t}) + n(e^{i\omega t} + e^{-i\omega t})) \\ &= \frac{\Delta_x^2}{2} ((n+1)2 \cos \omega t + 2n \cos \omega t) = \Delta_x^2 (2n+1) \cos \omega t. \end{aligned}$$

Problem 5

(i) We want to show

$$\hat{D}^\dagger(\alpha) \hat{a} \hat{D}(\alpha) = \exp(\alpha^* \hat{a} - \alpha \hat{a}^\dagger) \hat{a} \exp(\alpha \hat{a}^\dagger - \alpha^* \hat{a}) = \hat{a} + \alpha.$$

Recall the Hadamard operator lemma

$$e^A B e^{-A} = \sum_{n=0}^{\infty} \frac{[A, [\dots, [A, B] \dots]]}{n!}.$$

Take A to be the argument of the first exponential and $B = \hat{a}$. Then using the fact that $\hat{a}$ commutes with itself,

$$[A, B] = \alpha^* \hat{a} \hat{a} - \alpha \hat{a}^\dagger \hat{a} - \alpha^* \hat{a} \hat{a} + \alpha \hat{a} \hat{a}^\dagger = \alpha (\hat{a} \hat{a}^\dagger - \hat{a}^\dagger \hat{a}) = \alpha [\hat{a}, \hat{a}^\dagger] = \alpha 1.$$

Then note that $[A, \alpha 1] = 0$, so all higher order terms vanish. We are left with the 0th and 1st terms

$$e^A B e^{-A} = B + [A, B].$$

Plugging in our definition of A and B yields

$$\hat{D}^\dagger(\alpha) \hat{a} \hat{D}(\alpha) = \hat{a} + \alpha 1.$$

(ii) We want to show

$$\hat{D}(\alpha)\hat{a}\hat{D}^\dagger(\alpha) = \exp\left(\alpha\hat{a}^\dagger - \alpha^*\hat{a}\right)\hat{a}\exp\left(\alpha^*\hat{a} - \alpha\hat{a}^\dagger\right) = \hat{a} - \alpha.$$

Appealing once again to the Hadamart operator lemma, define

$$A = \alpha\hat{a}^\dagger - \alpha^*\hat{a}, \quad B = \hat{a}.$$

As before,

$$[A, B] = \left(\alpha\hat{a}^\dagger - \alpha^*\hat{a}\right)\hat{a} - \hat{a}\left(\alpha\hat{a}^\dagger - \alpha^*\hat{a}\right) = \alpha\left(\hat{a}^\dagger\hat{a} - \hat{a}\hat{a}^\dagger\right) = -\alpha[\hat{a}, \hat{a}^\dagger] = -\alpha 1.$$

Higher order terms in the expansion once again vanish, leaving us with

$$e^A B e^{-A} = B + [A, B] = \hat{a} - \alpha.$$

(iii) Conjugating both sides of the identity (i) yields

$$\hat{D}^\dagger(\alpha)\hat{a}^\dagger\hat{D}(\alpha) = (\hat{a} + \alpha 1)^\dagger = \hat{a}^\dagger + \alpha^* 1.$$

(iv) Conjugating both sides of the identity (ii) yields

$$\hat{D}(\alpha)\hat{a}^\dagger\hat{D}^\dagger(\alpha) = (\hat{a} - \alpha 1)^\dagger = \hat{a}^\dagger - \alpha^* 1.$$